

[EECS 304] Control I

Sheet (8) State Space Representation

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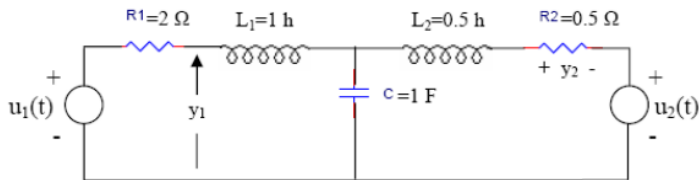
Overview

- **State:** The minimum information or the minimum number of variables to completely describe the system.
- State space representation (SSR) is a new method to model the system in the time domain, suitable for multivariable systems, convenient for computers and accounts for the initial conditions.
- Order of differential equation = number of state variables (number of x)
- SSR equations:
 - ▶ **State equation:** $\dot{\underline{x}} = A\underline{x} + B\underline{u}$
 - ▶ **Output equation:** $\underline{y} = C\underline{x} + D\underline{u}$
- A: state matrix, B: input matrix, C: output matrix
- $T.F. = C[sI - A]^{-1}B + D$
- Weighting function = $\ell^{-1}\{T.F.\}$

Problem 1

Problem 1

For the following networks write down the state equations and output equations and get the corresponding state space representation.



Problem 1

Solution

In this problem, we have 3 states

$$x_1 = I_1, \quad x_2 = V_c, \quad x_3 = I_2,$$

DEs:

$$v_{L_1} = L_1 \frac{di_1}{dt} = L_1 \frac{dx_1}{dt} = L_1 \dot{x}_1 \rightarrow (1)$$

$$v_{L_2} = L_2 \frac{di_2}{dt} = L_2 \frac{dx_3}{dt} = L_2 \dot{x}_3 \rightarrow (2)$$

$$I_{C_1} = C \frac{dv_c}{dt} = C \frac{dx_2}{dt} = C \dot{x}_2 \rightarrow (3)$$

Output Relations:

$$u_1 = I_1 R_1 + y_1 \rightarrow y_1 = u_1 - I_1 R_1 = -R_1 x_1 + u_1$$

$$y_2 = I_2 R_2 = R_2 x_3$$

Problem 1

Solution

State Equations:

$$L_1 \dot{x}_1 = v_{L_1} = y_1 - v_c = -R_1 x_1 + u_1 - x_2$$

$$\therefore \dot{x}_1 = -\frac{R_1}{L_1} x_1 - \frac{1}{L_1} x_2 + \frac{1}{L_1} u_1 \rightarrow (I)$$

$$C \dot{x}_2 = I_{C_1} = I_1 - I_2 = x_1 - x_3$$

$$\therefore \dot{x}_2 = \frac{1}{C} x_1 - \frac{1}{C} x_3 \rightarrow (II)$$

$$L_2 \dot{x}_3 = v_{L_2} = v_c - y_2 - u_2 = x_2 - R_2 x_3 - u_2$$

$$\therefore \dot{x}_3 = \frac{1}{L_2} x_2 - \frac{R_2}{L_2} x_3 - \frac{1}{L_2} u_2 \rightarrow (III)$$

Problem 1

Solution

State Space Representation:

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} -\frac{R_1}{L_1} & -\frac{1}{L_1} & 0 \\ \frac{1}{C} & 0 & -\frac{1}{C} \\ 0 & \frac{1}{L_2} & -\frac{R_2}{L_2} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} \frac{1}{L_1} & 0 \\ 0 & 0 \\ 0 & -\frac{1}{L_2} \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$$

$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} -R_1 & 0 & 0 \\ 0 & 0 & R_2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$$

Problem 1

Solution

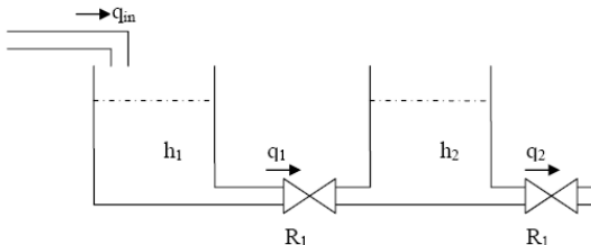
State Space Representation:

$$A_{3 \times 3} = \begin{bmatrix} -\frac{R_1}{L_1} & -\frac{1}{L_1} & 0 \\ \frac{1}{C} & 0 & -\frac{1}{C} \\ 0 & \frac{1}{L_2} & -\frac{R_2}{L_2} \end{bmatrix} \quad B_{3 \times 2} = \begin{bmatrix} \frac{1}{L_1} & 0 \\ 0 & 0 \\ 0 & -\frac{1}{L_2} \end{bmatrix}$$
$$C_{2 \times 3} = \begin{bmatrix} -R_1 & 0 & 0 \\ 0 & 0 & R_2 \end{bmatrix} \quad D_{2 \times 2} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$$

Problem 2

Problem 2

For the interacting two tank system shown below derive a suitable state space representation if the system has two outputs, the first output is the liquid level in the second tank and the second output is the output flow rate from the second tank.



Problem 2

Solution

DEs:

$$q_i - q_1 = A_1 \frac{dh_1}{dt}$$

$$q_1 - q_2 = A_2 \frac{dh_2}{dt}$$

We choose the states: $x_1 = h_1$, $x_2 = h_2$

$$\therefore q_i - q_1 = A_1 \dot{x}_1$$

$$\therefore q_1 - q_2 = A_2 \dot{x}_2$$

Output Equations:

$$y_1 = h_2 = x_2$$

$$y_2 = q_2 = \frac{h_2}{R_2} = \frac{1}{R_2} x_2$$

Problem 2

Solution

State Equations:

$$A_1 \dot{x}_1 = q_i - q_1$$

$$\therefore A_1 \dot{x}_1 = u - \frac{h_1 - h_2}{R_1} = u - \frac{x_1 - x_2}{R_1} = -\frac{1}{R_1}x_1 + \frac{1}{R_2}x_2 + u$$

$$\therefore \dot{x}_1 = -\frac{1}{A_1 R_1}x_1 + \frac{1}{A_1 R_2}x_2 + \frac{1}{A_1}u \rightarrow (I)$$

$$A_2 \dot{x}_2 = q_1 - q_2$$

$$\therefore A_2 \dot{x}_2 = \frac{h_1 - h_2}{R_1} - \frac{h_2}{R_2} = \frac{x_1 - x_2}{R_1} - \frac{x_2}{R_2}$$

$$\therefore \dot{x}_2 = -\frac{1}{A_2 R_1}x_1 - \frac{1}{A_2}\left(\frac{1}{R_1} + \frac{1}{R_2}\right)x_2 \rightarrow (II)$$

Problem 2

Solution

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} -\frac{1}{A_1 R_1} & \frac{1}{A_1 R_1} \\ \frac{1}{A_2 R_1} & -\frac{1}{A_2} \left(\frac{1}{R_1} + \frac{1}{R_2} \right) \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} \frac{1}{A_1} \\ 0 \end{bmatrix} u$$

$$\therefore A_{2 \times 2} = \begin{bmatrix} -\frac{1}{A_1 R_1} & \frac{1}{A_1 R_1} \\ \frac{1}{A_2 R_1} & -\frac{1}{A_2} \left(\frac{1}{R_1} + \frac{1}{R_2} \right) \end{bmatrix} \quad B_{2 \times 1} = \begin{bmatrix} \frac{1}{A_1} \\ 0 \end{bmatrix}$$

Problem 2

Solution

Output Equations:

$$y_1 = x_2$$

$$y_2 = \frac{1}{R_2} x_2$$

$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 0 & \frac{1}{R_2} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$\therefore C_{2 \times 2} = \begin{bmatrix} 0 & 1 \\ 0 & \frac{1}{R_2} \end{bmatrix} \quad D = 0$$

Problem 3

Problem 3

For the system:

$$\dot{\underline{x}} = \begin{bmatrix} -5 & -1 \\ 3 & -1 \end{bmatrix} \underline{x} + \begin{bmatrix} 1 \\ 1 \end{bmatrix} u, \quad y = \begin{bmatrix} 1 & 2 \end{bmatrix} \underline{x}$$

Obtain the T.F. of the system

Problem 3

Solution

$$\therefore T.F. = C(sI - A)^{-1}B + D$$

$$\therefore (sI - A) = \begin{bmatrix} s & 0 \\ 0 & s \end{bmatrix} - \begin{bmatrix} -5 & -1 \\ 3 & -1 \end{bmatrix} = \begin{bmatrix} s+5 & 1 \\ -3 & s+1 \end{bmatrix}$$

$$\therefore (sI - A)^{-1} = \frac{1}{(s+5)(s+1) + 3} \begin{bmatrix} s+1 & -1 \\ 3 & s+5 \end{bmatrix}$$

$$\therefore (sI - A)^{-1} = \frac{1}{s^2 + 6s + 8} \begin{bmatrix} s+1 & -1 \\ 3 & s+5 \end{bmatrix}$$

$$\therefore T.F. = \begin{bmatrix} 1 & 2 \end{bmatrix} \left(\frac{1}{s^2 + 6s + 8} \begin{bmatrix} s+1 & -1 \\ 3 & s+5 \end{bmatrix} \right) \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\therefore T.F. = \frac{1}{s^2 + 6s + 8} \begin{bmatrix} 1 & 2 \end{bmatrix} \begin{bmatrix} s+1-1 \\ 3+s+5 \end{bmatrix}$$

Problem 3

Solution

$$\therefore T.F. = \frac{1}{s^2 + 6s + 8} \begin{bmatrix} 1 & 2 \end{bmatrix} \begin{bmatrix} s \\ s + 8 \end{bmatrix} = \frac{s + 2(s + 8)}{s^2 + 6s + 8}$$

$$\therefore T.F. = \frac{3s + 16}{s^2 + 6s + 8}$$

$\therefore$ weighting function $w(t) = \ell^{-1}\{T.F.\}$

$$\therefore w(t) = \ell^{-1}\left\{\frac{3s + 16}{s^2 + 6s + 8}\right\} = \ell^{-1}\left\{\frac{-2}{s + 4} + \frac{5}{s + 2}\right\}$$

$$\therefore w(t) = (5e^{-2t} - 2e^{-4t})u(t)$$

Problem 4

Problem 4

Find a phase state space representation for the following systems:

a
$$\frac{Y(s)}{U(s)} = \frac{1}{(s+1)(s+3)}$$

b
$$\frac{Y(s)}{U(s)} = \frac{1}{2s^4 + s^3 + s^2 + 27s + 50}$$

c
$$\frac{Y(s)}{U(s)} = \frac{10s + 1}{2s^3 + s^2 + s + 10}$$

d
$$\ddot{y} + 5\ddot{y} + 3\dot{y} + y = 2\dot{f}(t) + f(t)$$

Problem 4

Solution

a)

$$T.F. = \frac{1}{s^2 + 4s + 3}$$

Using Phase SSR:

$$\begin{aligned}\dot{\underline{x}} &= \begin{bmatrix} 0 & 1 \\ -3 & -4 \end{bmatrix} \underline{x} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u \\ y &= \begin{bmatrix} 1 & 0 \end{bmatrix} \underline{x}\end{aligned}$$

Problem 4

Solution

b)

$$\frac{Y(s)}{U(s)} = \frac{1}{2s^4 + s^3 + s^2 + 27s + 50}$$

Using phase SSR, we should make T.F. in the standard form first $\rightarrow$
"coefficient of s^n should be 1"

$$\therefore \frac{Y(s)}{U(s)} = \frac{0.5}{s^4 + 0.5s^3 + 0.5s^2 + 13.5s + 25}$$

$$\dot{\underline{x}} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -25 & -13.5 & -0.5 & -0.5 \end{bmatrix} \underline{x} + \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} u$$

$$y = [0.5 \quad 0 \quad 0 \quad 0] \underline{x}$$

Problem 4

Solution

c)

$$\frac{Y(s)}{U(s)} = \frac{10s + 1}{2s^3 + s^2 + s + 10}$$

Using phase SSR, we should make T.F. in the standard form first $\rightarrow$
"coefficient of s^n should be 1"

$$\therefore \frac{Y(s)}{U(s)} = \frac{5s + 0.5}{s^3 + 0.5s^2 + 0.5s + 5}$$

$$\dot{\underline{x}} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -5 & -0.5 & -0.5 \end{bmatrix} \underline{x} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u$$
$$y = [0.5 \quad 1 \quad 0] \underline{x}$$

Problem 4

Solution

d)

$$\ddot{y} + 5\dot{y} + 3y = 2\dot{f}(t) + f(t)$$

Taking Laplace

$$\therefore (s^3 + 5s^2 + 3s + 1)Y(s) = (2s + 1)F(s)$$

$$\therefore \frac{Y(s)}{F(s)} = \frac{2s + 1}{s^3 + 5s^2 + 3s + 1}$$

$$\dot{\underline{x}} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -1 & -3 & -5 \end{bmatrix} \underline{x} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u$$

$$y = [1 \quad 2 \quad 0] \underline{x}$$

Problem 5

Problem 5

A certain system is described by the following set of DEs:

$$\ddot{y}_1 + \dot{y}_1 + 2y_1 - 2y_2 = u_1$$

$$\ddot{y}_2 - y_1 + y_2 = u_2$$

- a Write the state and output equations of the system.
- b Find the transfer function matrix between the outputs and inputs.

Problem 5

Solution

In this problem, we notice that we have 2 i/ps u_1, u_2 and 2 o/ps y_1, y_2 . So, we will choose the states as following:

$$x_1 = y_1, \quad x_2 = \dot{y}_1, \quad x_3 = y_2, \quad x_4 = \dot{y}_2$$

$$\therefore \dot{x}_1 = \dot{y}_1 = x_2$$

$$\therefore \dot{x}_3 = \dot{y}_2 = x_4$$

$$\therefore \dot{x}_2 = \ddot{y}_1 = u_1 - \dot{y}_1 - 2y_1 + 2y_2 = u_1 - x_2 - 2x_1 + 2x_3$$

$$\therefore \dot{x}_4 = \ddot{y}_2 = u_2 + y_1 - y_2 = u_2 + x_1 - x_3$$

Problem 5

Solution

$$\therefore \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \\ \dot{x}_4 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ -2 & -1 & 2 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & -1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 1 & 0 \\ 0 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$$

$$\therefore \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}$$

$$\therefore TFM = C[sI - A]^{-1}B = [TFM]_{2 \times 2}$$

TFM: Transfer Function Matrix

Problem 5

Solution

$$\therefore (sI - A) = \begin{bmatrix} s & 0 & 0 & 0 \\ 0 & s & 0 & 0 \\ 0 & 0 & s & 0 \\ 0 & 0 & 0 & s \end{bmatrix} - \begin{bmatrix} 0 & 1 & 0 & 0 \\ -2 & -1 & 2 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & -1 & 0 \end{bmatrix} = \begin{bmatrix} s & -1 & 0 & 0 \\ 2 & s+1 & -2 & 0 \\ 0 & 0 & s & -1 \\ -1 & 0 & 1 & s \end{bmatrix}$$

$$\therefore (sI - A)^{-1} = \frac{1}{s^4 + s^3 + 3s^2 + s} \begin{bmatrix} M_{11} & -M_{12} & M_{13} & -M_{14} \\ -M_{21} & M_{22} & -M_{23} & M_{24} \\ M_{31} & -M_{32} & M_{33} & -M_{34} \\ -M_{41} & M_{42} & -M_{43} & M_{44} \end{bmatrix}^T$$

$$\therefore T.F. = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \frac{1}{|sI - A|} \begin{bmatrix} M_{11} & -M_{12} & M_{13} & -M_{14} \\ -M_{21} & M_{22} & -M_{23} & M_{24} \\ M_{31} & -M_{32} & M_{33} & -M_{34} \\ -M_{41} & M_{42} & -M_{43} & M_{44} \end{bmatrix}^T \begin{bmatrix} 0 & 0 \\ 1 & 0 \\ 0 & 0 \\ 0 & 1 \end{bmatrix}$$

Problem 5

Solution

$$\therefore T.F. = \frac{1}{s^4 + s^3 + 3s^2 + s} \begin{bmatrix} M_{11} & -M_{21} & M_{31} & -M_{41} \\ M_{13} & -M_{23} & M_{33} & -M_{43} \end{bmatrix} \begin{bmatrix} 0 & 0 \\ 1 & 0 \\ 0 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\therefore T.F. = \frac{1}{s^4 + s^3 + 3s^2 + s} \begin{bmatrix} -M_{21} & -M_{41} \\ -M_{23} & -M_{43} \end{bmatrix}$$

$$\therefore T.F. = \frac{1}{s^4 + s^3 + 3s^2 + s} \begin{bmatrix} s^2 + 1 & 2 \\ 1 & s^2 + s + 2 \end{bmatrix}$$

$$\therefore T.F. = \begin{bmatrix} \frac{s^2 + 1}{s^4 + s^3 + 3s^2 + s} & \frac{2}{s^4 + s^3 + 3s^2 + s} \\ -\frac{1}{s^4 + s^3 + 3s^2 + s} & \frac{s^2 + s + 2}{s^4 + s^3 + 3s^2 + s} \end{bmatrix}$$

Thank you!

Questions?

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